

PROBLEMS SUBMIT CODE MY SUBMISSIONS STATUS STANDINGS CUSTOM INVOCATION

I. Count Paths Queries

time limit per test: 6 s
memory limit per test: 256 megabytes

You are given a graph with n vertices (numbered 1 through n) and m **directed** edges.

You need to answer q queries: given s, t, k , count paths consisting of k steps (edges) from vertex s to vertex t , modulo $10^9 + 7$. A path can visit the same vertex or edge multiple times.

Input

The first line contains three integers n, m and q ($1 \leq n, q \leq 200, 0 \leq m \leq n \cdot (n - 1)$) — respectively the number of vertices, edges and queries.

The following m lines describe edges. The i -th of these lines contains two integers a_i and b_i ($1 \leq a_i, b_i \leq n, a_i \neq b_i$), denoting a directed edge from a_i to b_i . There are no self-loops or multiple edges.

The following q lines describe queries. Each line contains three integers s, t and k ($1 \leq s, t \leq n, 1 \leq k \leq 10^9$) — the start and end of a path, and its length in edges.

Output

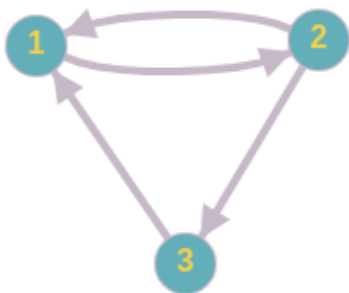
For each query, print a line with the answer modulo 1000000007.

Example

input	Copy
3 4 4 1 2 2 3 3 1 2 1 1 2 6 3 3 5 1 3 1 3 2 54	
output	Copy
2 1 0 922111	

Note

In the first sample test, we are given the following graph with $n = 3$ vertices and $m = 4$ edges.



In the first query, there are 2 paths of length $k = 6$ from vertex 1 to vertex 2:

- $1 \rightarrow 2 \rightarrow 3 \rightarrow 1 \rightarrow 2 \rightarrow 1 \rightarrow 2$
- $1 \rightarrow 2 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 1 \rightarrow 2$

Matrix Exponentiation

Finished

Practice



→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
336357441	Aug/31/2025 12:40	Happy New Year!
336356583	Aug/31/2025 12:34	Stuck in sleigh traffic on flight 13 TL13
336356287	Aug/31/2025 12:32	Stuck in sleigh traffic on flight 12 TL12

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